Bicol University
POLANGUI CAMPUS
Centro Occidental Polangui, Albay

IT112: WEB SYSTEMS and TECHNOLOGIE'S

Project Proposal
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BentaBoard

"From bawat benta to tunay na kita—track it right."

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BentaBoard

I. Introduction

BentaBoard is a **Small Seller Business Management System** is a simple yet powerful web-based business management tool designed specifically for small-scale sellers in the Philippines. Many micro-entrepreneurs, such as sari-sari store owners, market vendors, and online sellers, still rely on manual record-keeping methods like handwritten notebooks or mental calculations to track their business finances. This approach is highly prone to errors and makes it difficult to assess overall business performance. BentaBoard addresses this gap by providing an intuitive, easy-to-use system that automates financial tracking and profit calculation, eliminating the need for complicated accounting software.

II. System Purpose

The primary purpose of the BentaBoard is to help small sellers efficiently manage their daily business operations in one centralized platform. The system begins with a secure authentication process where users can register for an account and log in to access their personalized business dashboard. Once authenticated, the system provides tools for product management (adding, editing, and deleting items with selling prices), daily sales recording (logging what was sold, quantities, and revenue), expense tracking (operational costs such as rent, utilities, and transportation), and inventory purchase logging (recording when new stock is bought from suppliers). Most importantly, the system automatically computes the seller's profit on a daily, monthly, and yearly basis by subtracting total expenses and purchases from total sales revenue.

III. Target Users

- **Sari-sari store owners** – Neighborhood convenience store operators who need to track inventory and daily sales of various consumer goods.
- **Market vendors** – Sellers at public wet/dry markets who deal with multiple products, variable pricing, and daily restocking.
- **Online sellers** – Social media-based and e-commerce sellers who manage orders, track shipping costs, and need clear profit visibility.
- **Small food business operators** – Carinderia owners, food stall operators, and home-based food sellers who need to track ingredient purchases against daily sales.

IV. Key Features

1. User Registration and Authentication – New users can create an account by providing their username, email, password, full name, and role designation. The system securely stores user credentials with password hashing and validates user information during registration to prevent duplicate accounts.

2. Secure Login/Logout System – Registered users can log in using their username/email and password. The system validates credentials, creates secure sessions using Express Sessions or JWT tokens, and allows users to safely log out, terminating their active sessions.

3. Product Management – Sellers can add, edit, and delete products with their corresponding selling prices, categories, and current stock levels. This serves as the foundation for all transaction recording in the system.

4. Daily Sales Recording – The system allows sellers to log daily sales transactions by selecting a product, entering the quantity sold, and automatically computing the total amount received based on the defined selling price.

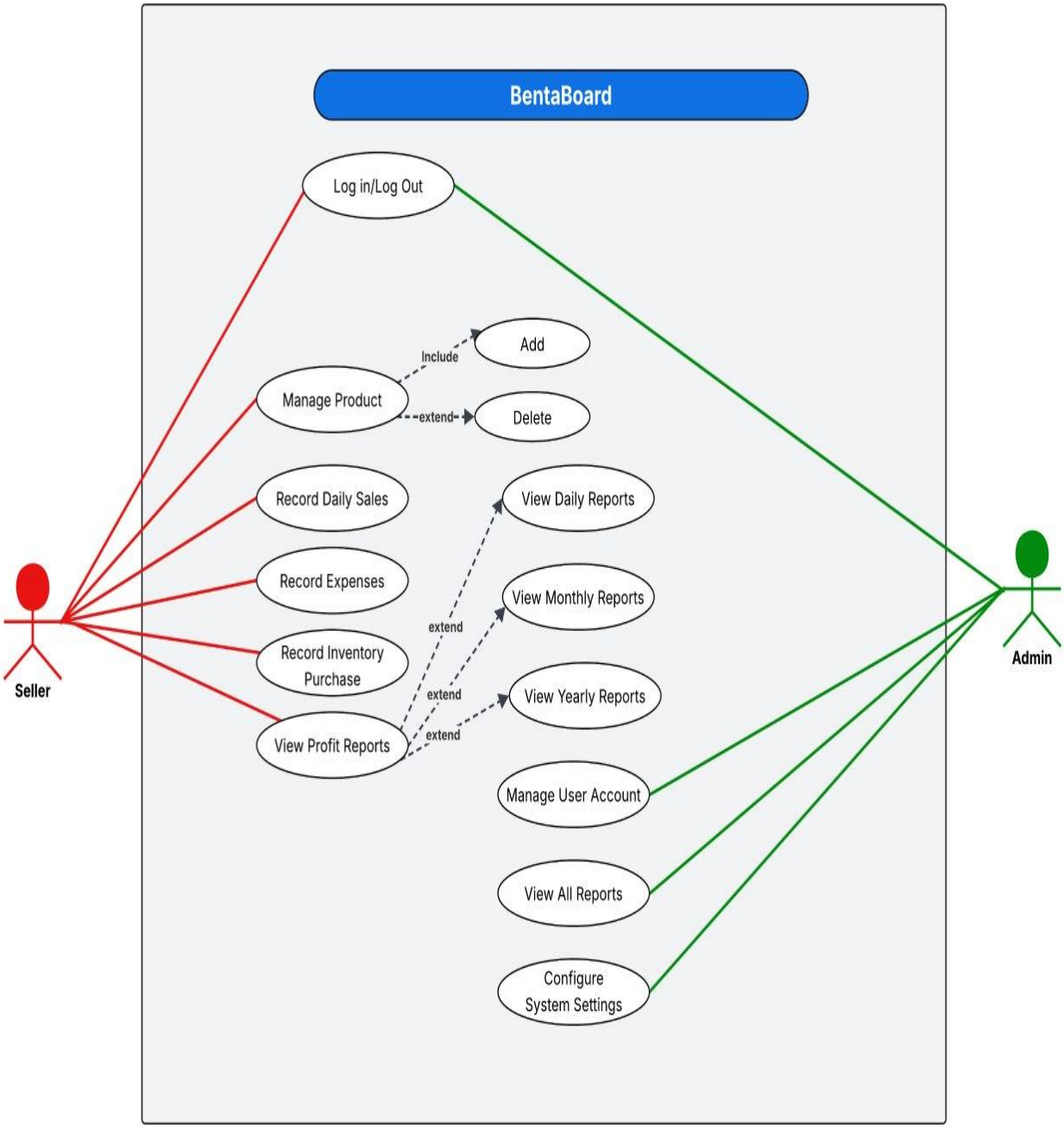
5. Expense Tracking – All operational expenses, such as the rent, utilities, transportation, supplies, and miscellaneous costs that can be recorded with proper categorization, dates, and descriptions for better financial visibility.

6. Inventory Purchase Logging – When sellers buy new stock from suppliers, they can log the purchase details including supplier name, quantity, unit cost, and total cost. The system automatically updates product stock levels accordingly.

7. Automated Profit Calculation – The system automatically computes daily, monthly, and yearly profit using the formula: Profit = Total Sales − (Total Expenses + Total Purchases). A visual dashboard provides a quick overview of the business's financial health.

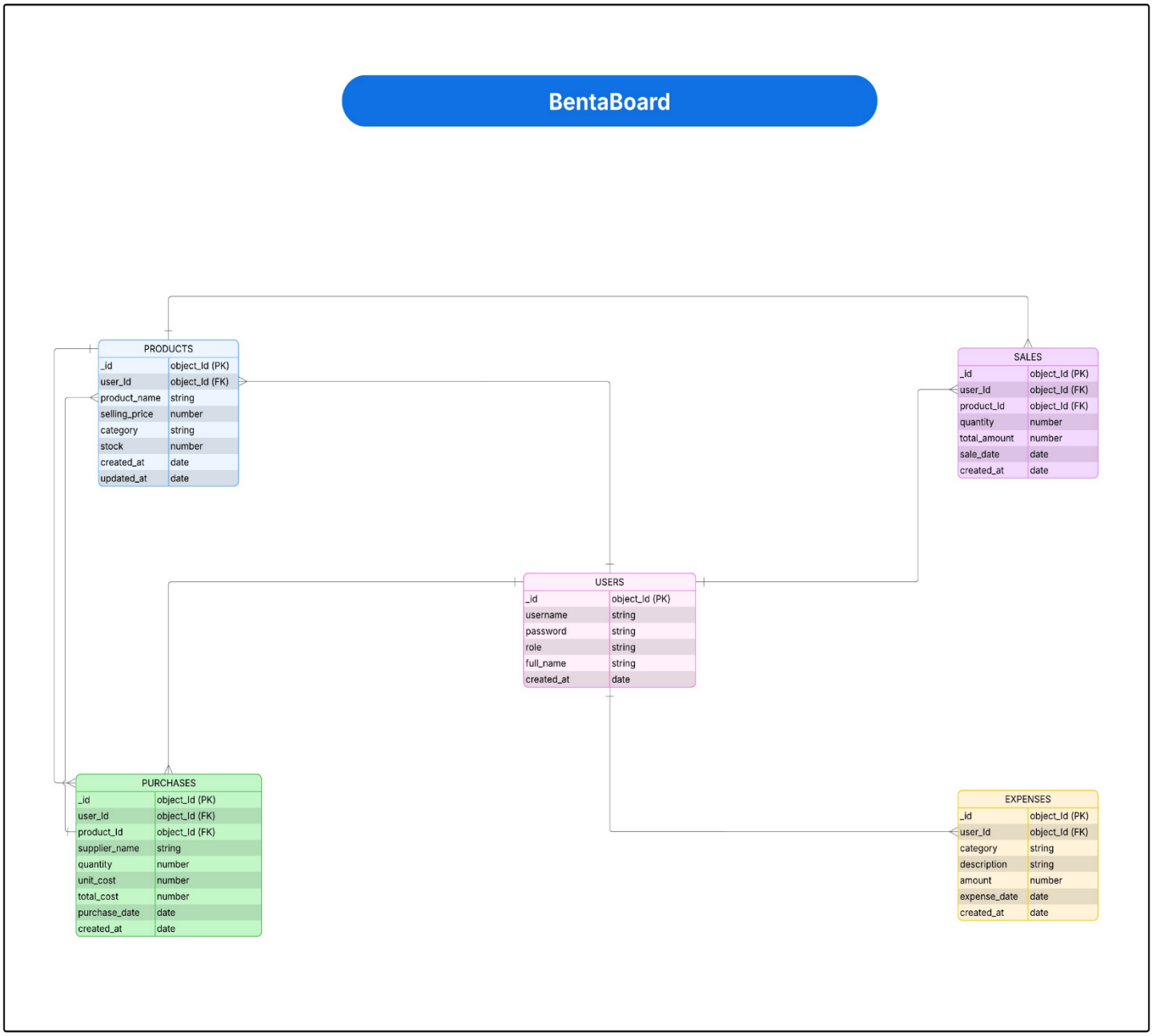
Use Case Diagram (UML)

The Use Case Diagram below illustrates how the two primary actors, Seller and Admin that interact with the BentaBoard system. Both actors must first register (if new) and log in to access the system. The Seller is the main user who performs day-to-day business operations including managing products, recording sales, logging expenses and purchases, and viewing profit reports and dashboard. The Admin handles system-level operations including user account management, viewing all reports, and configuring system settings. Both actors share the Register, Login, and Logout functionality.



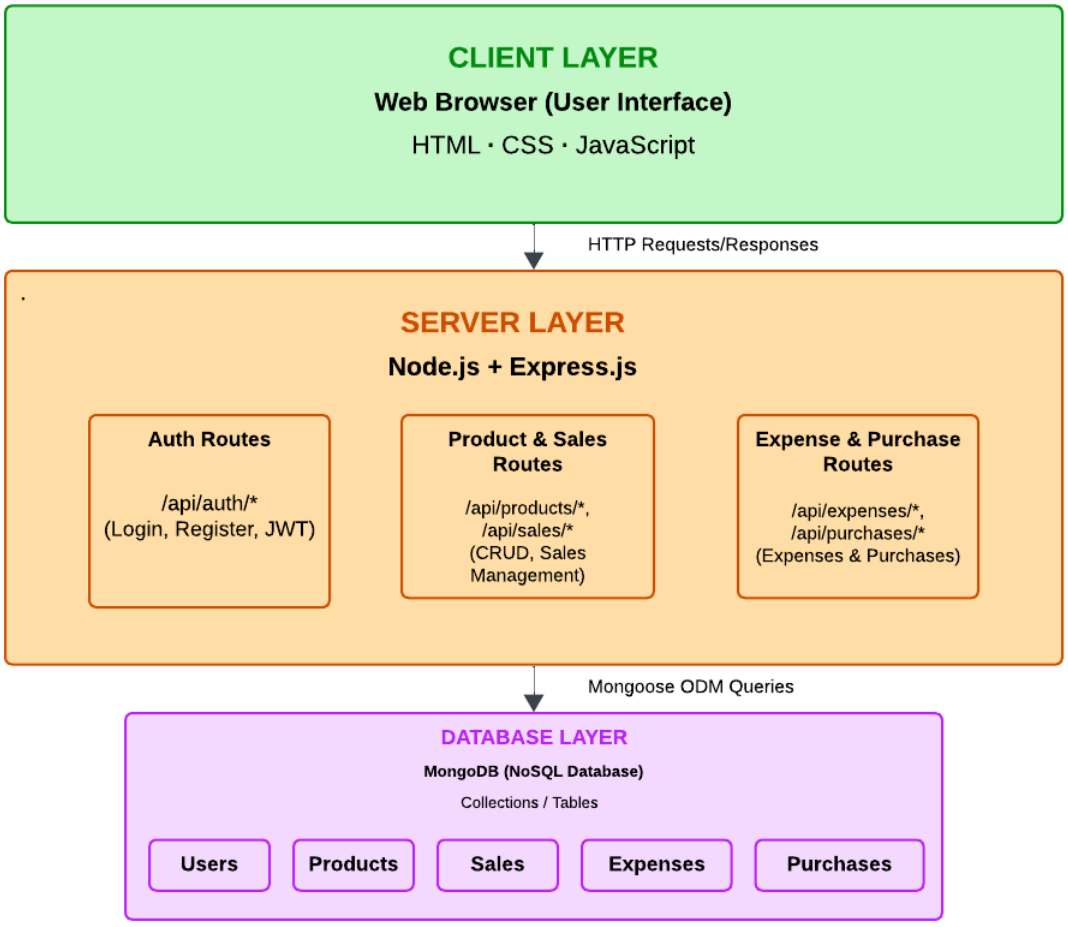
Entity-Relationship Diagram (ERD)

The ERD below represents the five main MongoDB collections used in BentaBoard, along with their attributes, data types, and inter-collection relationships. Primary keys (PK) and foreign key references (FK).



System Architecture Diagram

The System Architecture follows a three-tier web application design. The Client Layer handles the user interface through a standard web browser using HTML, CSS, and JavaScript. The Server Layer runs on Node.js with Express.js, handling API routing and business logic through dedicated route modules for authentication (registration/login/logout), product/sales management, and expense/purchase processing. The Database Layer uses MongoDB as the NoSQL database, with Mongoose ODM managing the communication between the server and database. No external APIs are required.



Proponents:

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